NavalForge Concept

Preliminary conceptual design report

10 m patrol launch — NF-DEMO-PATROL-10M — Revision P1

Engineering gate: CONDITIONAL — preliminary results with active engineering reservations

Technical notice: Preliminary engineering result. It does not replace the responsible engineer, formal approval, validated software, detailed structural analysis or verified input data.

Main indicators

Indicator	Value	Unit
Adherence	100.0	%
Displacement	7915.66	kg
Maximum speed	38.00	kn
Required installed power	299.13	kW
Range	510.29	nmi
Corrected GM	0.627	m
Freeboard	0.880	m

Requirement matrix

ID	Requirement	Actual	Result
REQ10-LOA	Overall length shall not exceed 10.50 m	10.0 m	PASS
REQ10-BEAM	Beam shall not exceed 3.30 m	3.1 m	PASS
REQ10-SPEED	Maximum speed shall be at least 35 kn	38.0 kn	PASS
REQ10-RANGE	Range shall be at least 170 nmi	510.2880704456275 nmi	PASS
REQ10-GM	Preliminary corrected GM shall exceed 0.45 m	0.6272697216930424 m	PASS
REQ10-FB	Preliminary freeboard shall exceed 0.50 m	0.8802774910584624 m	PASS
REQ10-PAYLOAD	Payload shall be at least 500 kg	500.0 kg	PASS

Warnings and assumptions

• GZ curve uses a wall-sided preliminary estimate; validate with inclined geometry before statutory use.

• Partially filled tanks present; free-surface correction applied.

• Consumption must be replaced by verified engine/propulsor curves for contractual use.

• All project and price data are synthetic demonstrations.

• Waterjet package is represented by aggregate demonstrative data.

• Conceptual parametric hull used unless verified offsets are supplied.

• Demonstrative prices are synthetic and are not current quotations.

Selected alternatives

NF-ECO — NF-DEMO-PATROL-10M-V002

Lowest demonstrative cost among variants passing every mandatory requirement.

NF-BALANCED — NF-DEMO-PATROL-10M-V004

Highest auditable multicriteria balance of cost, performance, range, GM and risk.

NF-PERFORMANCE — NF-DEMO-PATROL-10M-V005

Highest combined performance and technical-margin screening among compliant variants.